OPEN SOURCE INTELLIGENCE (OSINT)

CTF Challenges

GROUP 2 MEMBERS & THEIR OBJECTIVES

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1. Understand OSINT
2. Understand the use of Sherlock
3. Understand the use of WayBack Machine
4. Understand the OSINT Framework
5. OSINT in Systems Vulnerability Assessment



OSNIT DEFINATION

- **OSINT**, or **Open-Source Intelligence**, is the practice of gathering, analyzing, and disseminating information from publicly available sources to address specific intelligence requirements.
- In the context of **Capture The Flag (CTF)** competitions, OSINT, or Open Source Intelligence, refers to the practice of gathering and analyzing publicly available information to solve challenges and find hidden flags.

OSNIT TOOLS

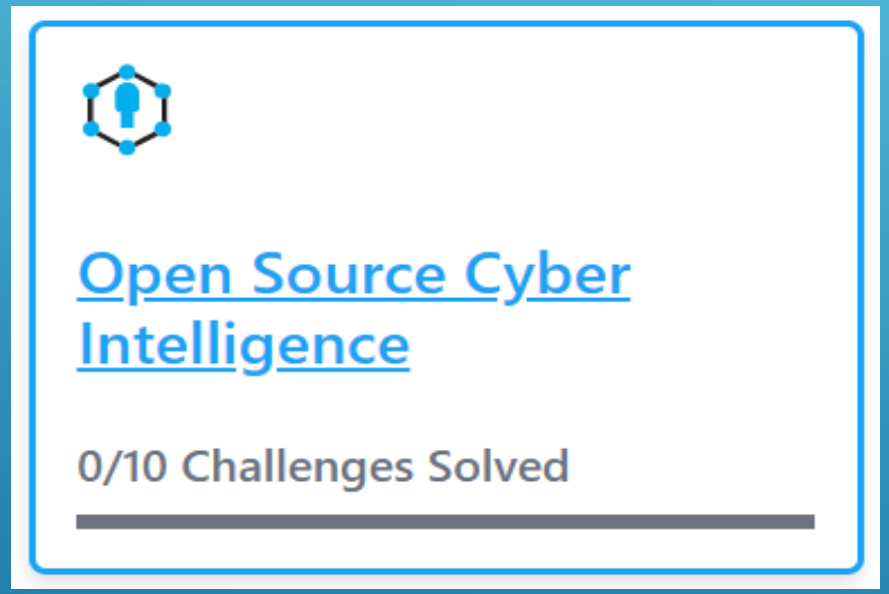
1. **Sherlock**

- This is a Linux command line tools that is used to search through different platforms(e.g.: twitter, Facebook) for the presence of the account of the username that you will provide to it.
- Install on Linux terminal with **sudo apt install sherlock** or
- Download the tool on **GitHub**

2. **Way Back Machine**

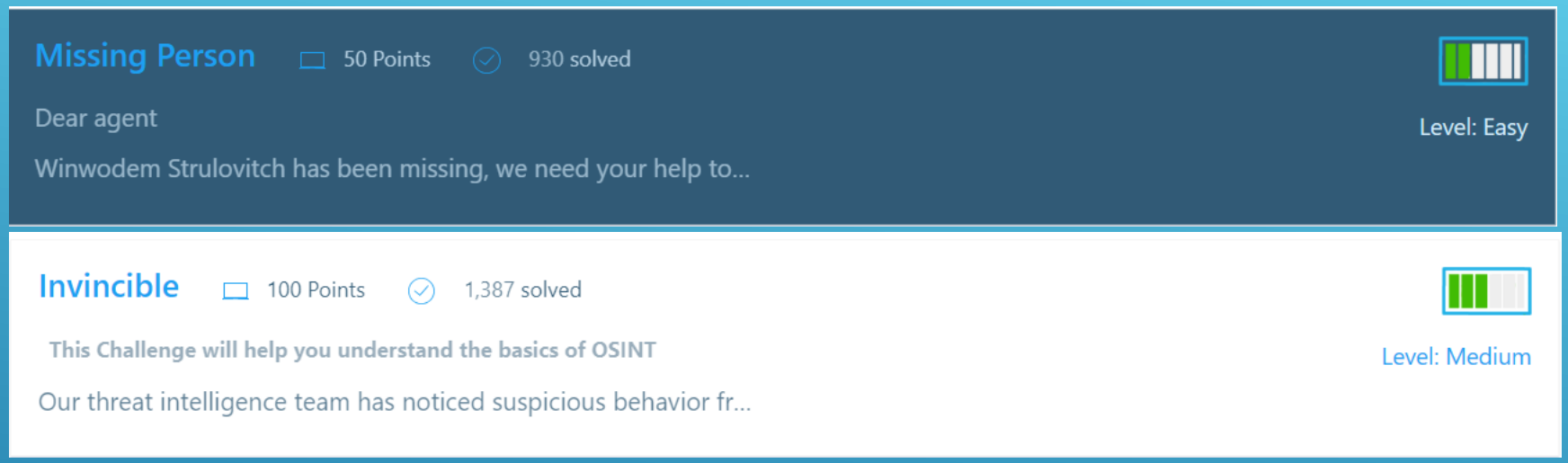
- The WayBack Machine is an online tool that allows users to access archived versions of websites from the past.
- It provides a timeline of snapshots to analyze the evolution and growth of a website.
- It can as well as retrieve information that may no longer be available on the current website.

- Sign up for **CyberTalents**
- Practice **Open Source Cyber Intelligence**



CYBER TALENTS

Challenges



CHALLENGE

- Downloading the attached PDF

```
┌──[sokojames@parrot]─[~/Desktop/CTF/OSNIT]
└─$wget https://hubchallenges.s3.eu-west-1.amazonaws.com/osint/winwodem.pdf
--2025-03-16 10:47:10--  https://hubchallenges.s3.eu-west-1.amazonaws.com/osint/
winwodem.pdf
Resolving hubchallenges.s3.eu-west-1.amazonaws.com (hubchallenges.s3.eu-west-1.a
mazonaws.com)... 3.5.69.108, 52.218.90.144, 52.218.37.200, ...
Connecting to hubchallenges.s3.eu-west-1.amazonaws.com (hubchallenges.s3.eu-west
-1.amazonaws.com)|3.5.69.108|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 160219 (156K) [application/pdf]
Saving to: 'winwodem.pdf'

winwodem.pdf        100%[===================>] 156.46K   165KB/s    in 0.9s

2025-03-16 10:47:13 (165 KB/s) - 'winwodem.pdf' saved [160219/160219]

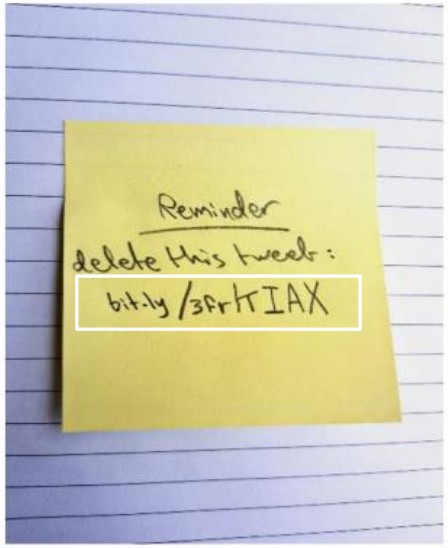
┌──[sokojames@parrot]─[~/Desktop/CTF/OSNIT]
└─$ls
winwodem.pdf
┌──[sokojames@parrot]─[~/Desktop/CTF/OSNIT]
└─$open winwodem.pdf
```

MISSING PERSON



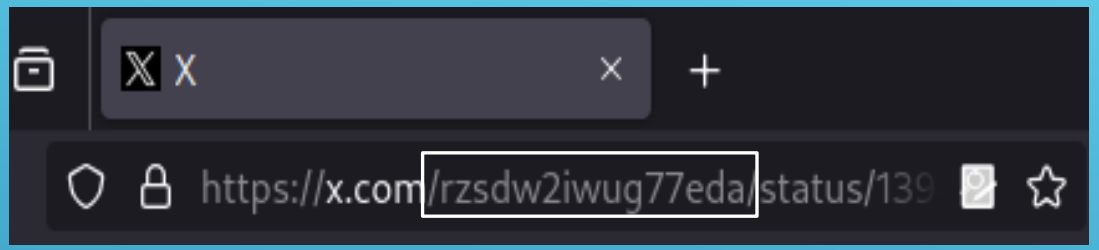
First Name: Winwodem
Last Name: Strulovitch
Nationality: American
Job: Accountant
Age: 30
Height: 185

This sticky note was found in a notebook in his room:



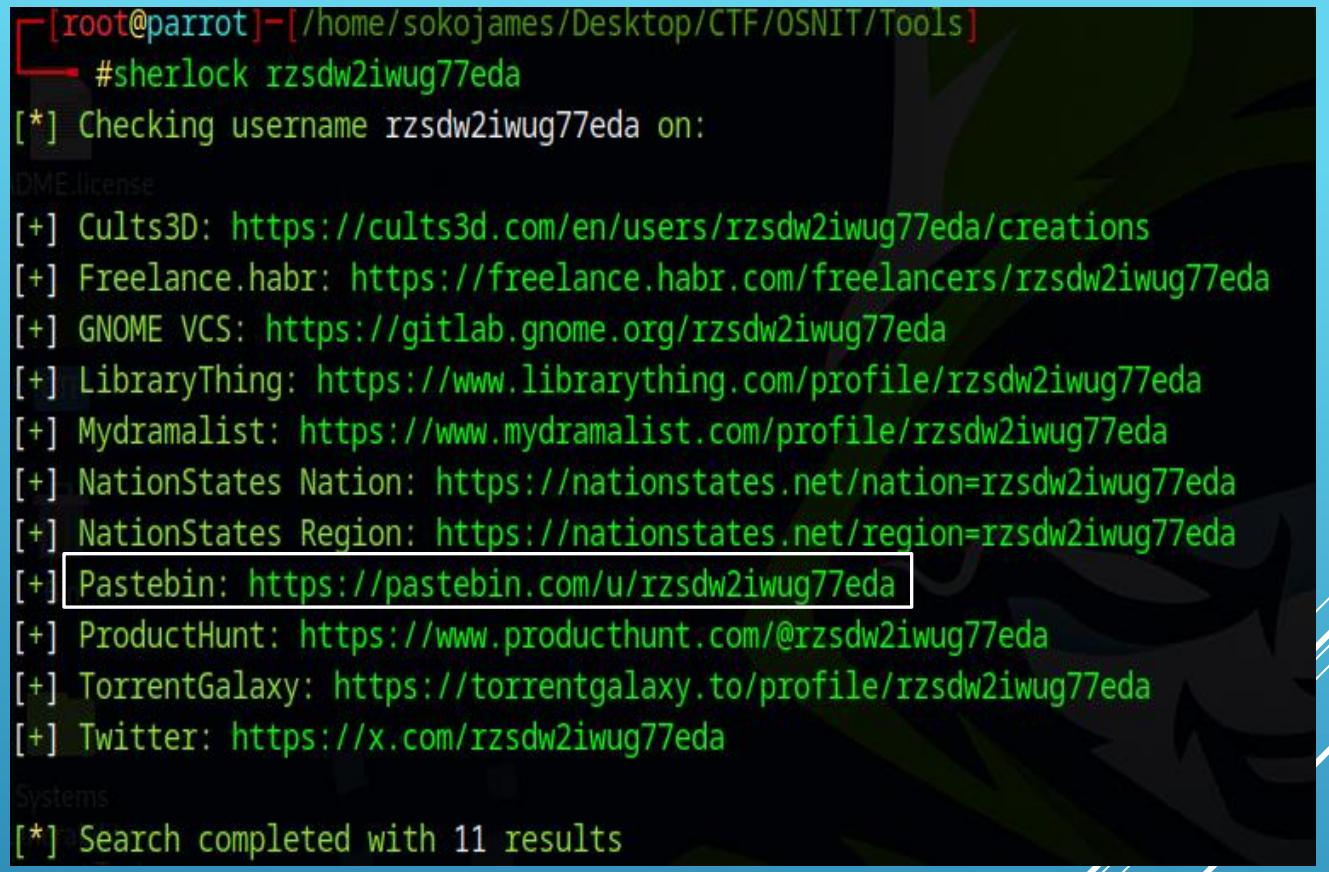
MISSING PERSON (CONTINUES…)

- **NOTE:** We are left with this link https://bitly.com/3frKIAX from the sticky note.
- This link redirects us to X (Twitter)
- Notice the username from the URL



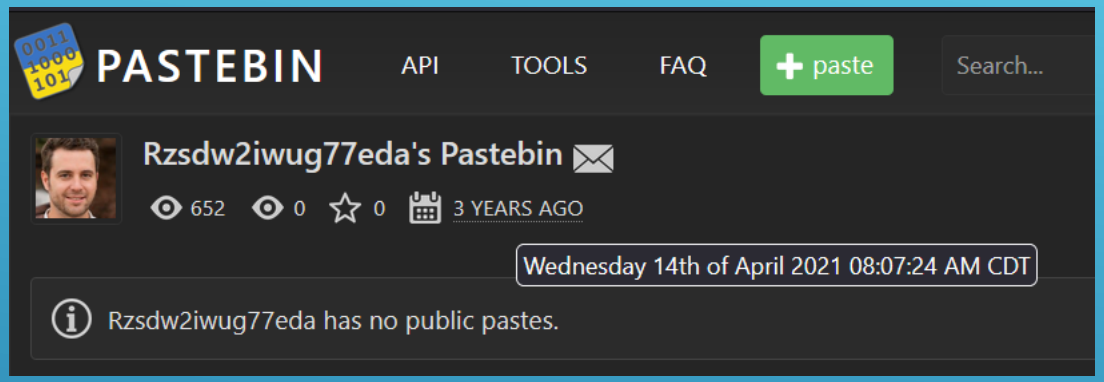
MISSING PERSON (CONTINUES…)

- We use the sherlock tool to find this missing user.
- **sherlock rzsdw2iwug77eda**



MISSING PERSON (CONTINUES…)

- Notice the date using the **Pastebin URL**



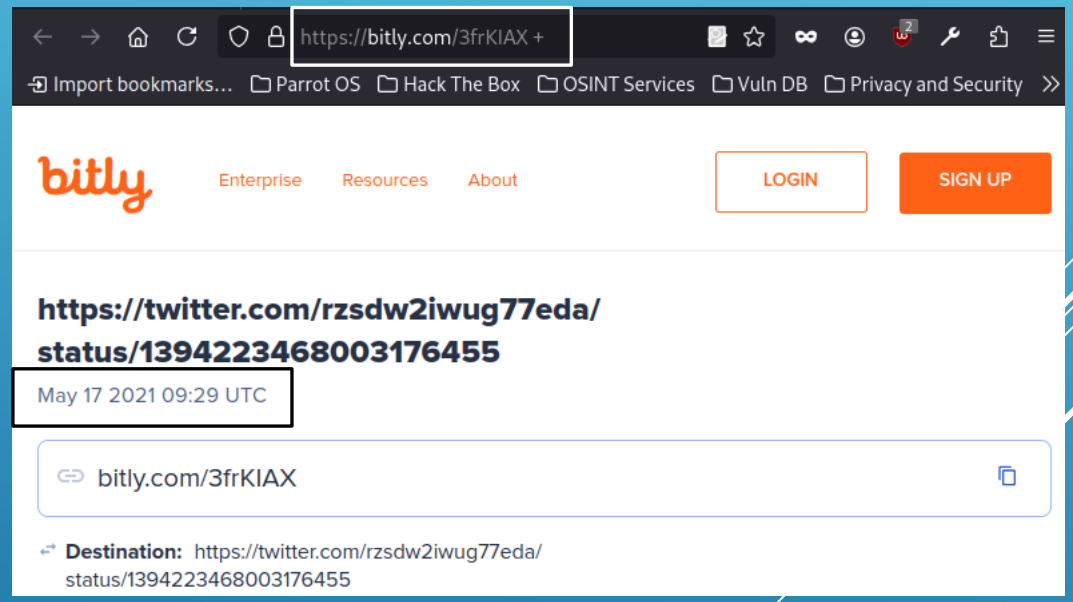
- flag{dd-mm-yyyy_hh:mm}
- flag{14-04-2021_08:07}
- This flag is incorrect, because we've identified the timestamp is he last seen online not the day he went missing.

MISSING PERSON (CONTINUES…)

- Remember a sticky note was left in his room and bitly link on it.
- Lets now find out when that link was created.

- The + at the end of a Bitly link will:

1. Help view the full original URL
2. Help us click on the analytics (if public)
3. Help us see when the link was created



MISSING PERSON (CONTINUES…)

- flag{dd-mm-yyyy_hh:mm}
- flag{17-05-2021_09:29}

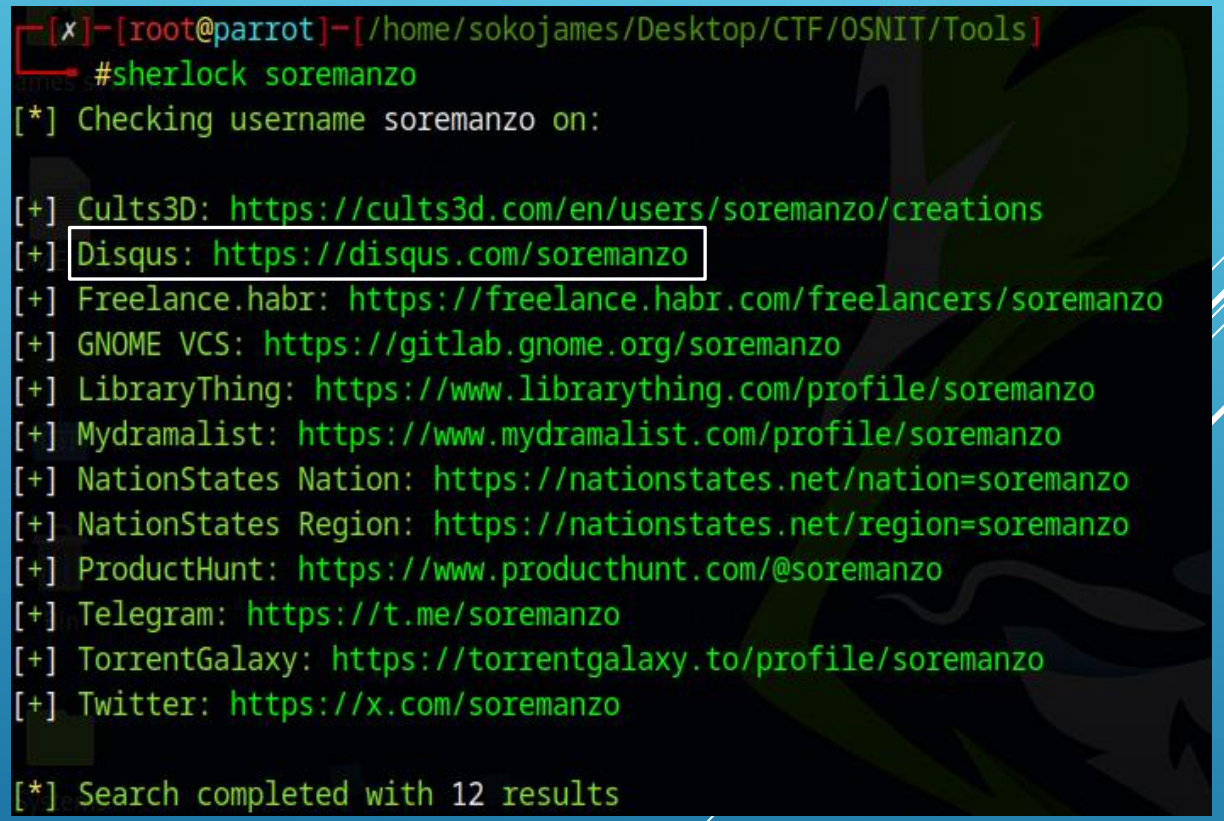
- Finally we've got our flag.



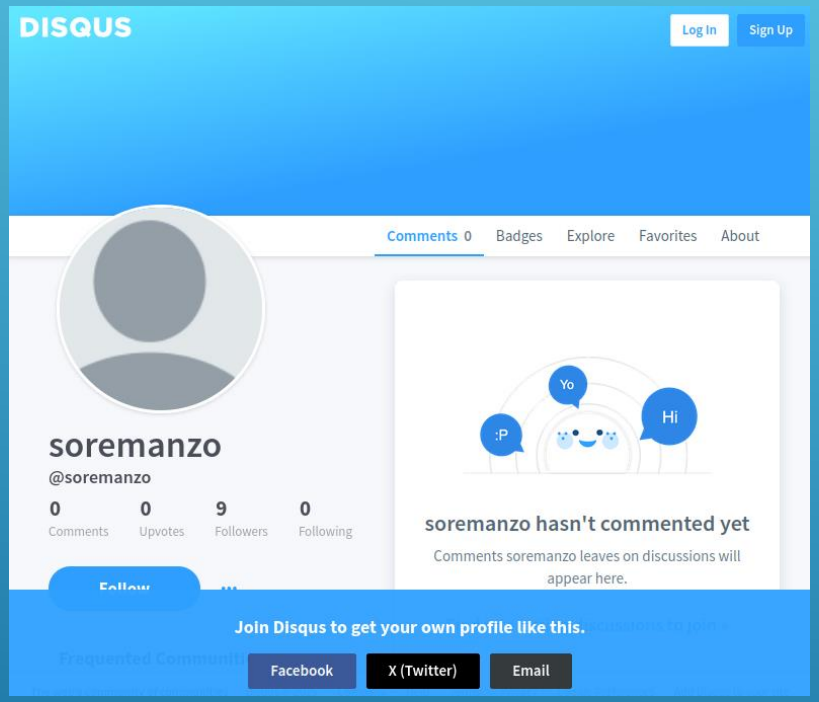
INVINCIBLE

- Our threat intelligence team has noticed suspicious behavior from the user "**soremanzo**" , Can you trace him and find what he is trying to hide.

- Based on the knowledge we've gained in the first challenge, will use **sherlock** tool to search for this user.



- It seems like the user deleted the flag

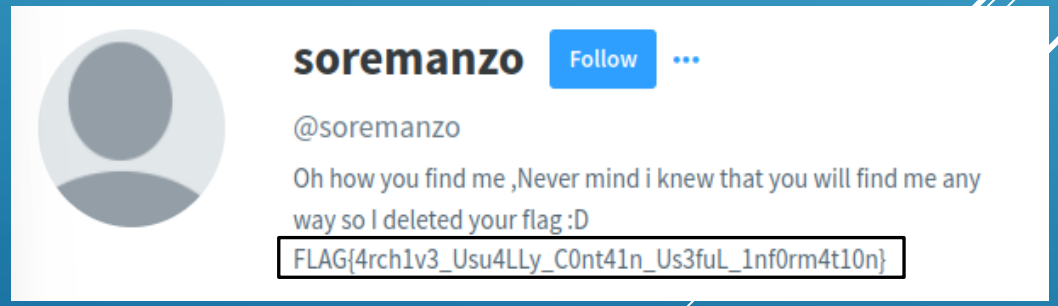


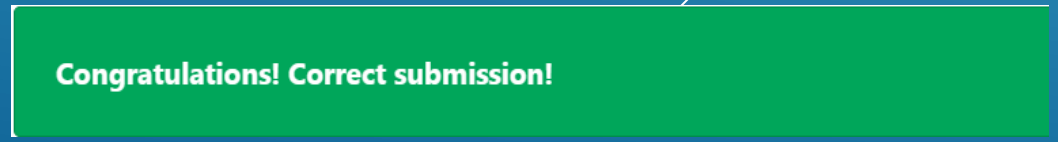
INVINCIBLE (CONTINUES…)

- Use **Way Back Machine** to see the changes that were made on this site.









SYSTEMS VULNERABILITY ASSESSMENT

Tool	Purpose	Use in Vulnerability Assessment
WayBack Machine	Archives old versions of websites	1. Find exposed credentials, admin portals, or sensitive data. 2. Analyze past security flaws and outdated technologies. 3. Compare security changes over time
Sherlock	Finds usernames across platforms	1. Identify exposed usernames for credential attacks. 2. Detect impersonation and phishing risks. 3. Support reconnaissance for penetration testing.



OSINT FRAME WORK

- **OSINT framework** focused on gathering information from free tools or resources.
- The intention is to help people find free OSINT resources.

END OF PRESENTATION